



Bursa Uludağ University
Faculty of Engineering
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Software Testing and Automation

Automation Exercise – Product and Price Verification Test

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AutomationExercise – Product and Price Verification Test

1. Introduction

In this assignment an automated test scenario was developed to verify the correctness of product and price information on an e-commerce web application. The test was implemented using Selenium WebDriver in a Kali Linux environment.

The main objective of this study is to ensure that the product details displayed on the products page remain consistent with the information shown in the shopping cart after user interactions.

2. Test Environment

The test was conducted in the following environment:

- Operating System: Kali Linux
- Programming Language: Python
- Automation Tool: Selenium WebDriver
- Browser: Chromium
- Driver: ChromeDriver

This environment enables reliable interaction with web elements and supports automated UI testing.

3. Test Scenario

The implemented test scenario consists of the following steps:

1. Navigate to the AutomationExercise products page
2. Retrieve all available products
3. Randomly select multiple products
4. Extract product names and prices
5. Add selected products to the cart
6. Navigate to the cart page
7. Extract cart data including:
 - a. Product name
 - b. Unit price
 - c. Quantity
 - d. Total price
8. Compare selected product data with cart data

This scenario ensures that the system maintains data consistency throughout the user journey.

4. Validation Criteria

The following validation checks were performed:

- The number of products added to the cart matches the selected products
- Product names are identical between the product page and the cart
- Product prices are consistent
- Quantity values are correct (expected value: 1)
- Total price for each product is correctly calculated
- Overall cart total matches the expected total

Assertions were used to automatically verify these conditions.

5. Results

The test execution produced successful results:

- All randomly selected products were added to the cart without errors
- Product names and prices were correctly transferred
- Quantity and total price values were accurate
- All validation checks passed successfully

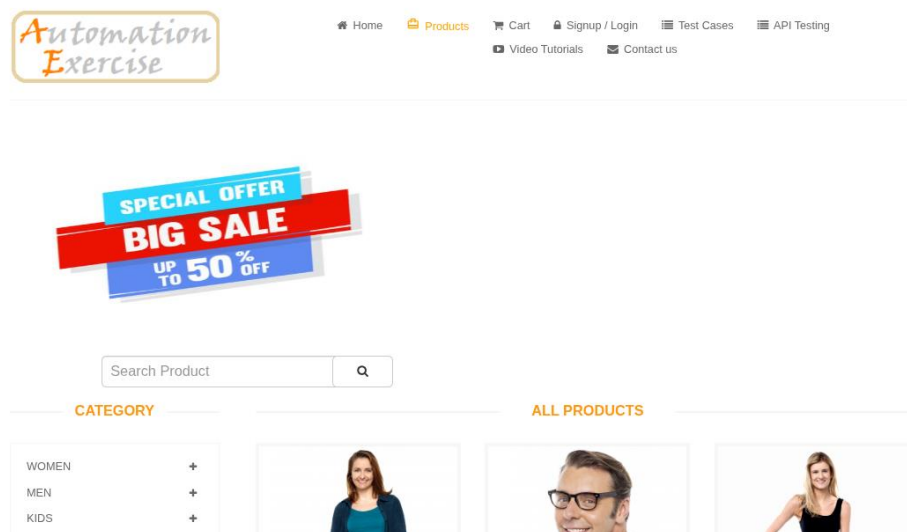


Figure 1. Products Page

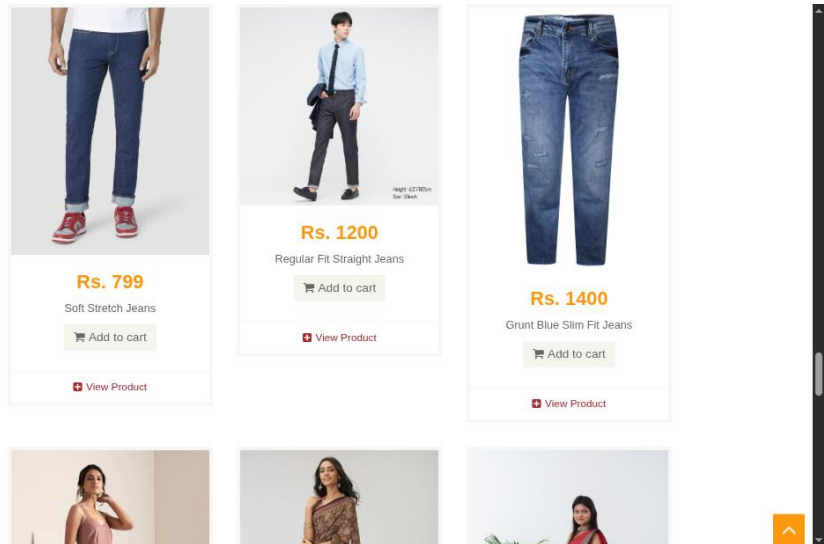


Figure 2. Selected Products to Cart

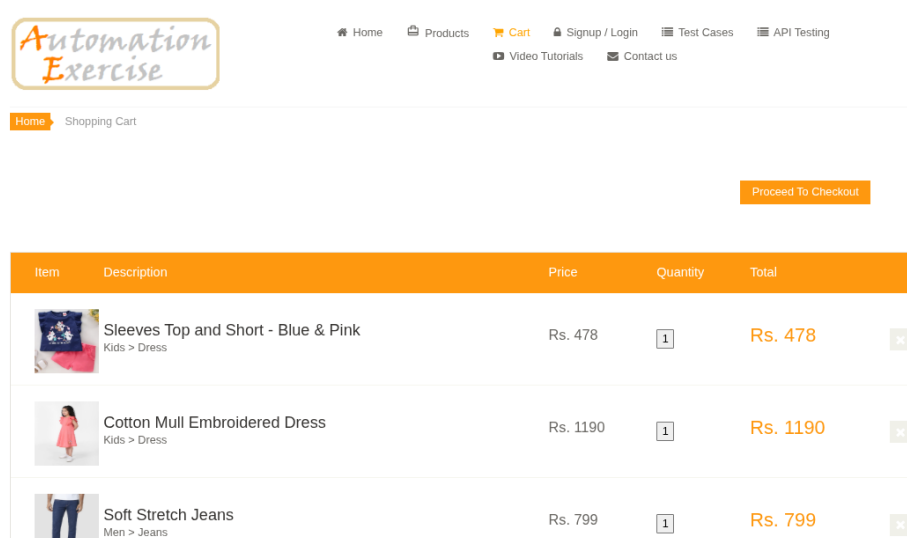
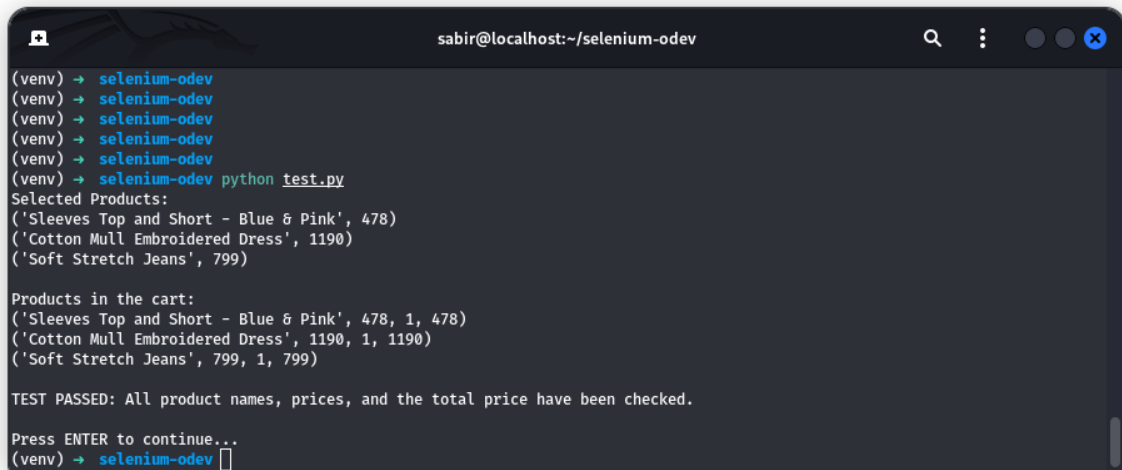


Figure 3. Cart Page with Product Details

A terminal window titled 'sabir@localhost:~/selenium-odev' showing the execution of a Python test script. The terminal displays the following output:

```
(venv) → selenium-odev
(venv) → selenium-odev
(venv) → selenium-odev
(venv) → selenium-odev
(venv) → selenium-odev
(venv) → selenium-odev python test.py
Selected Products:
('Sleeves Top and Short - Blue & Pink', 478)
('Cotton Mull Embroidered Dress', 1190)
('Soft Stretch Jeans', 799)

Products in the cart:
('Sleeves Top and Short - Blue & Pink', 478, 1, 478)
('Cotton Mull Embroidered Dress', 1190, 1, 1190)
('Soft Stretch Jeans', 799, 1, 799)

TEST PASSED: All product names, prices, and the total price have been checked.

Press ENTER to continue...
(venv) → selenium-odev
```

Figure 4. Successful Test Execution Output

6. Conclusion

This assignment demonstrates the effectiveness of automated testing in validating data consistency within web applications. By using Selenium WebDriver, the testing process becomes faster, more reliable, and less dependent on manual intervention.

The implemented test successfully verified that product information, including names, prices, and total calculations, remains accurate throughout the user workflow.